

README

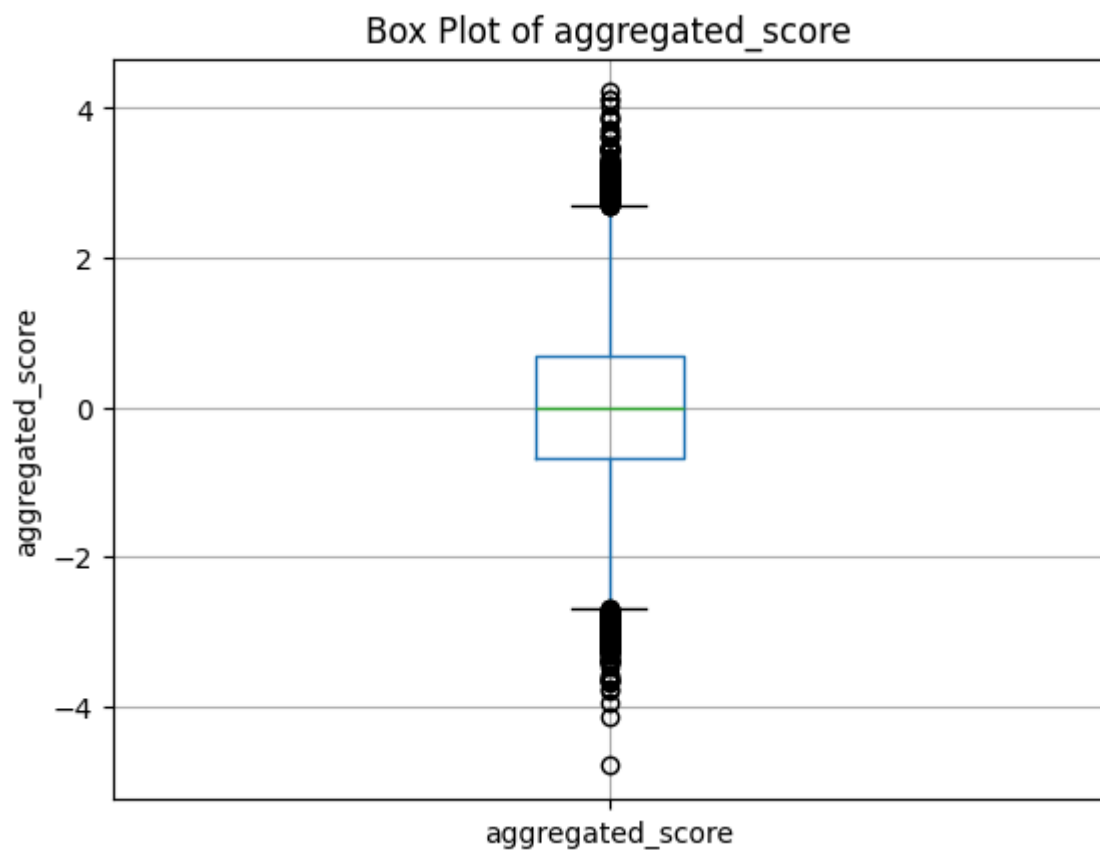
1. EXPLORAREA DATELOR

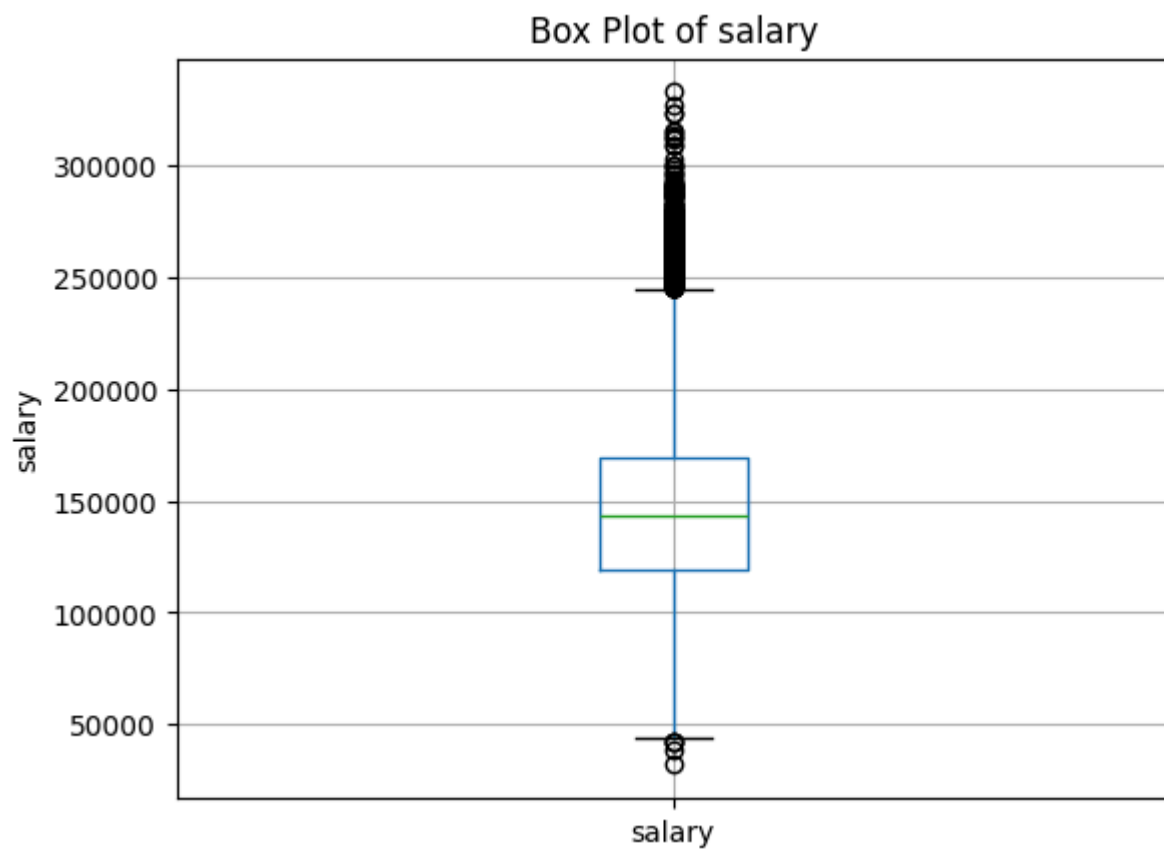
analiza tipului de atribut si a plajei de valori

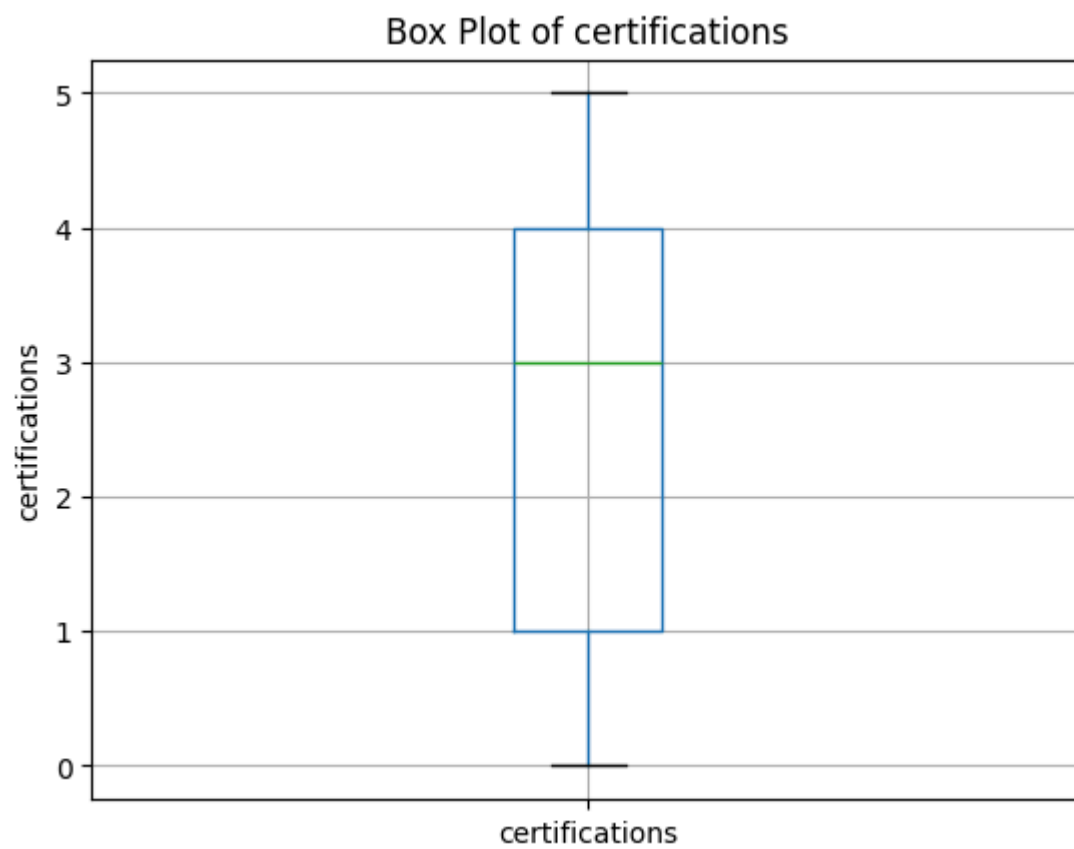
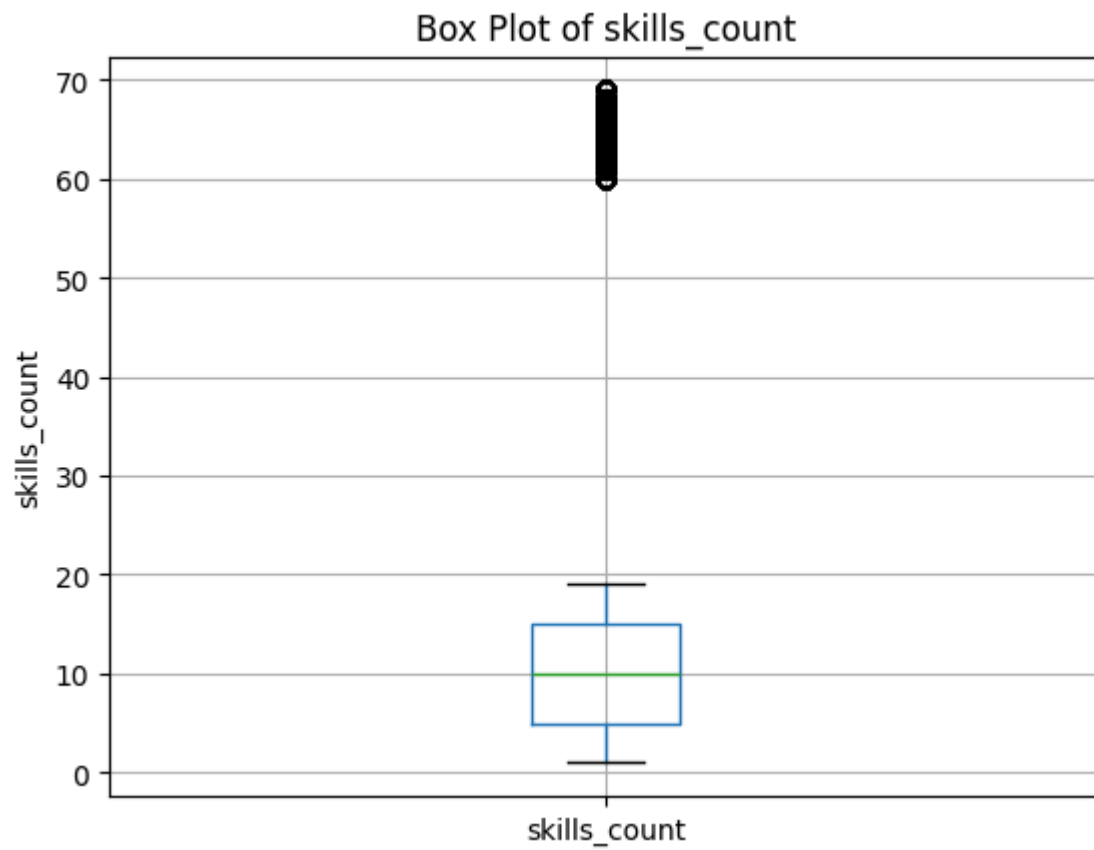
atribute numerice:

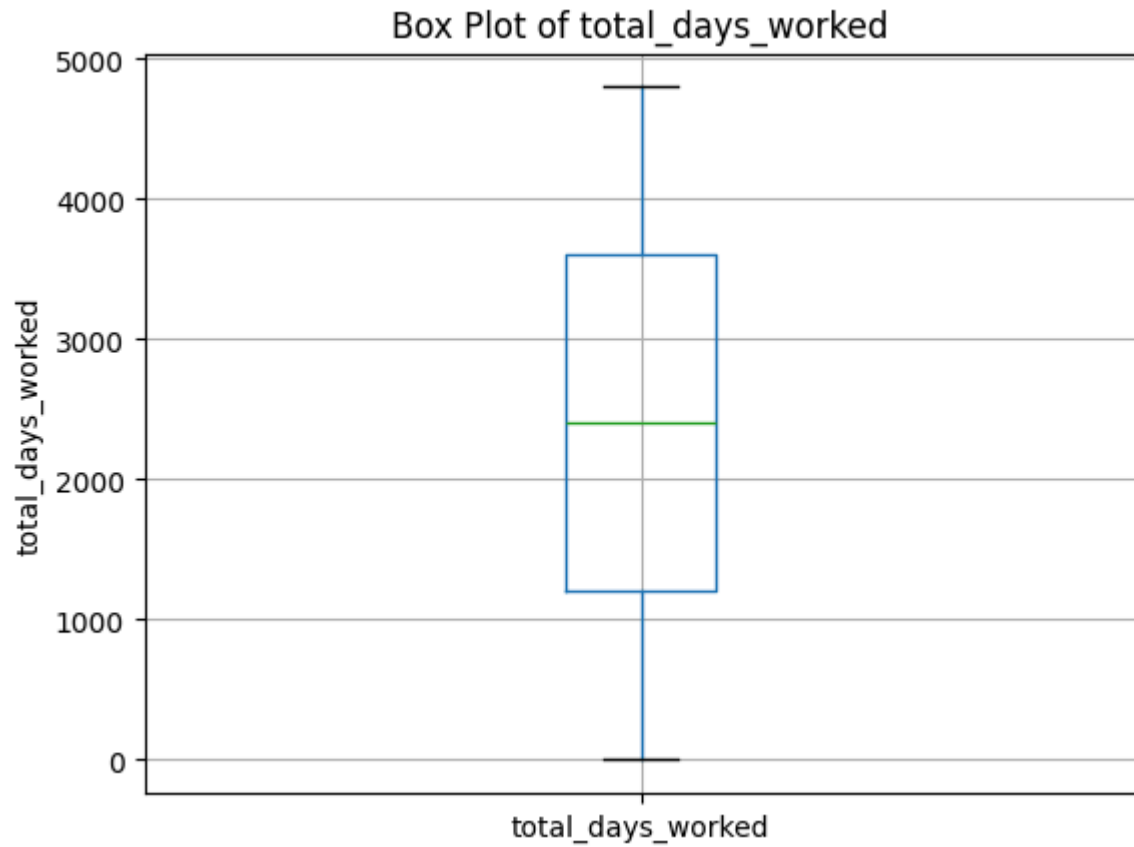
	aggrega ted_scor e	salary	experien ce_year s	skills _coun t	certifi cation s	total_day s_worke d
co un t	64000.00 0000	64000. 00000 0	64000.00 0000	64000 .0000 00	64000. 00000 0	64000.00 0000
m ea n	0.000929	14568 3.5658 44	9.973391	11.604 047	2.5111 41	2393.508 219
st d	0.999869	37384. 75571 5	6.050682	10.73 7497	1.7098 11	1452.044 678
mi n	-4.79690 2	31867. 00000 0	0.000000	1.000 000	0.0000 00	-1.00000 0
25 %	-0.67375 7	119209 .75000 0	5.000000	5.000 000	1.0000 00	1200.000 000

50 %	0.001059	143299.000000	10.000000	10.000000	3.000000	2400.000000
75 %	0.678273	169582.250000	15.000000	15.000000	4.000000	3600.000000
max	4.217828	333046.000000	20.000000	69.000000	5.000000	4800.000000





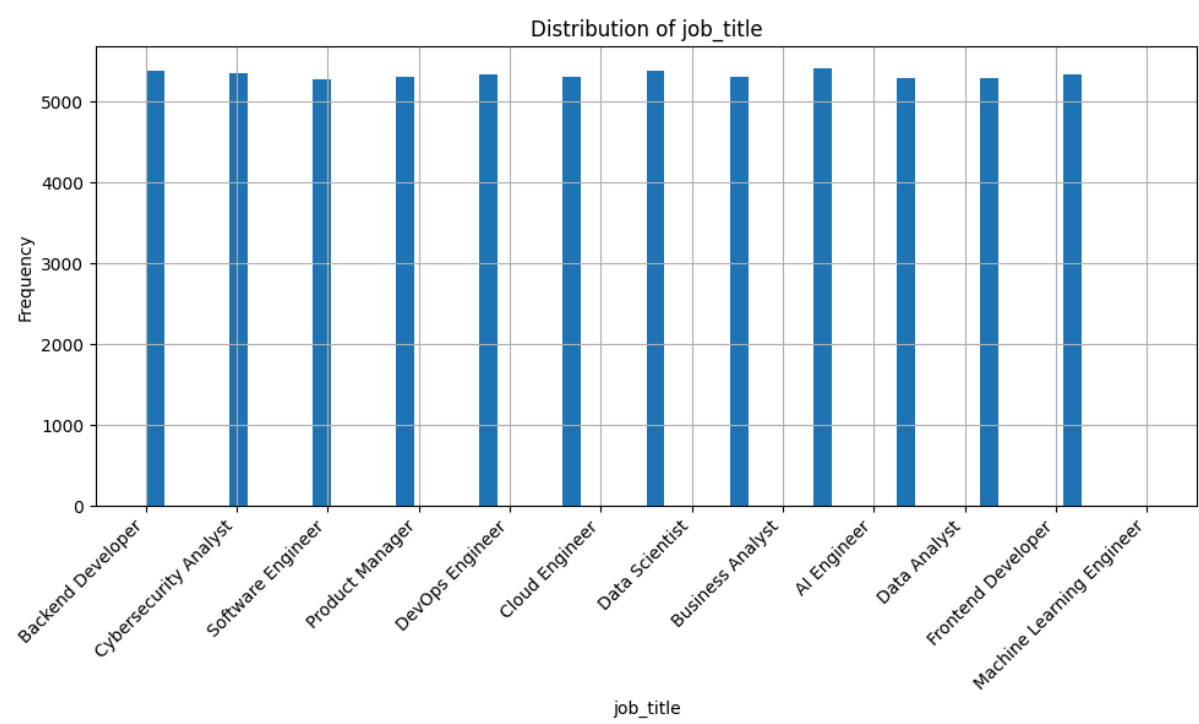


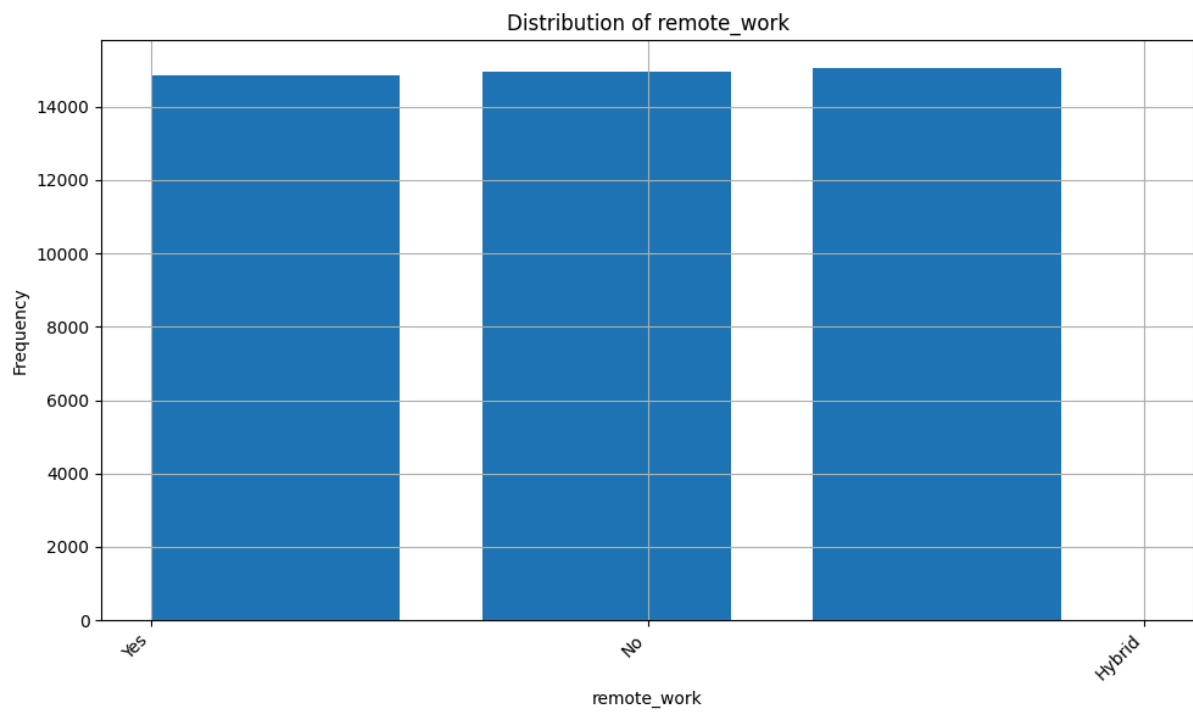
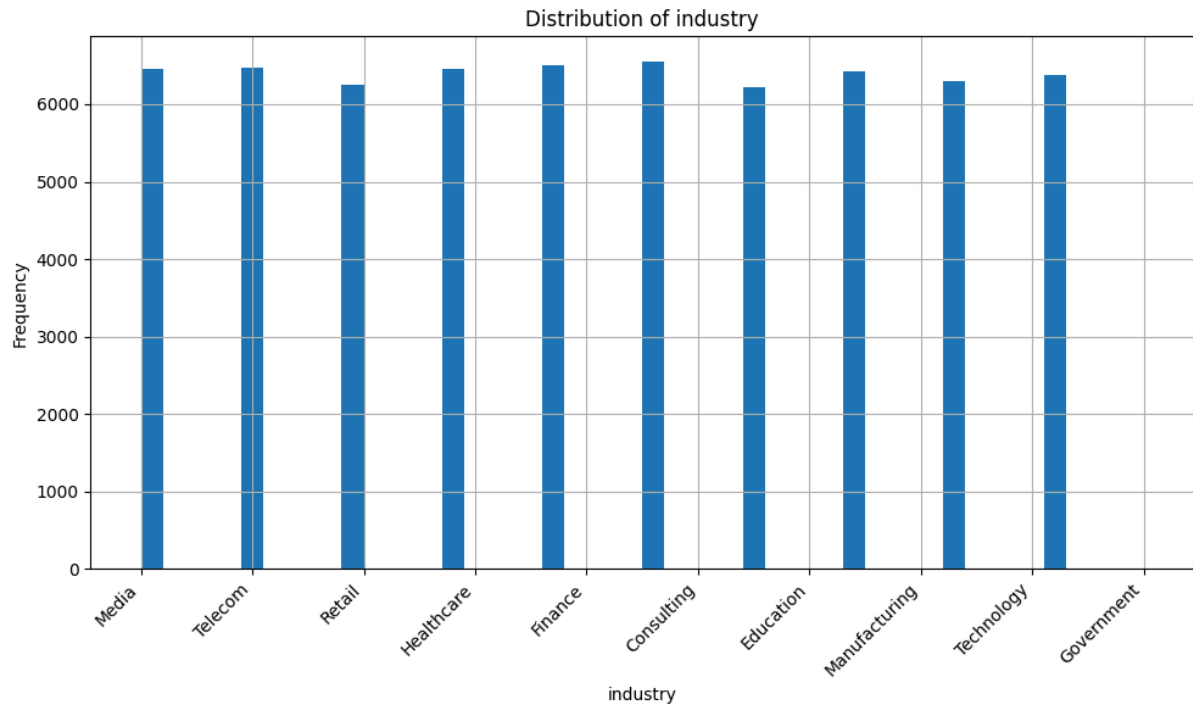


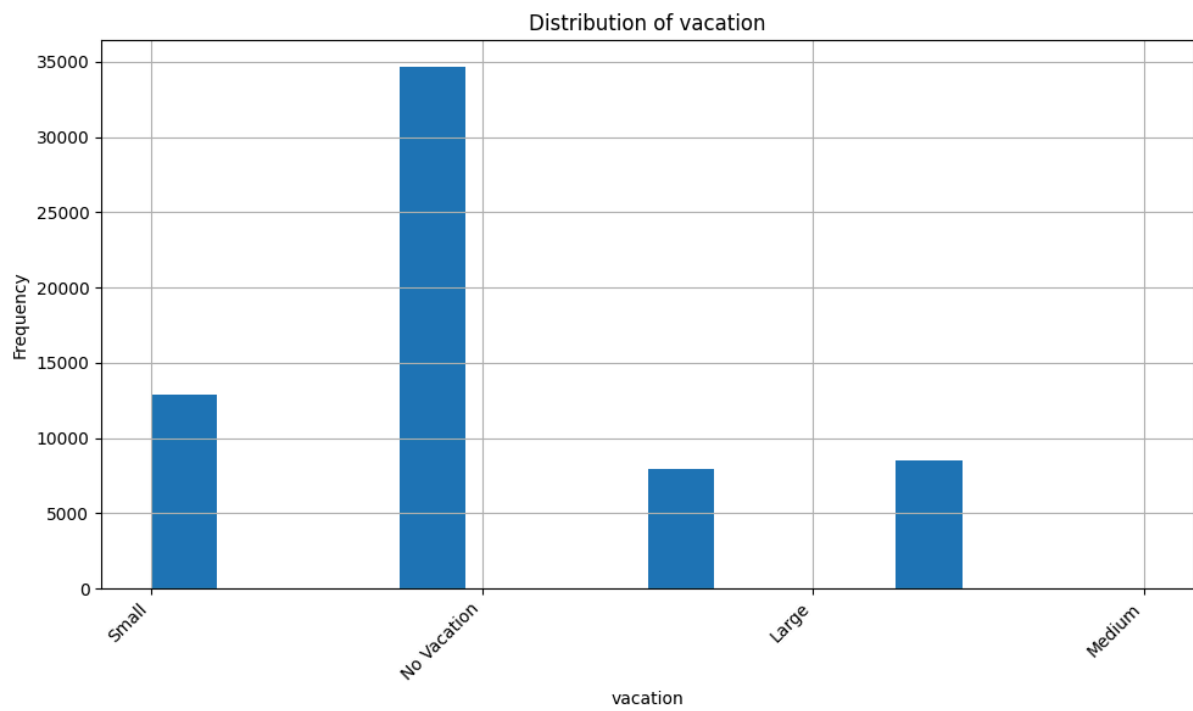
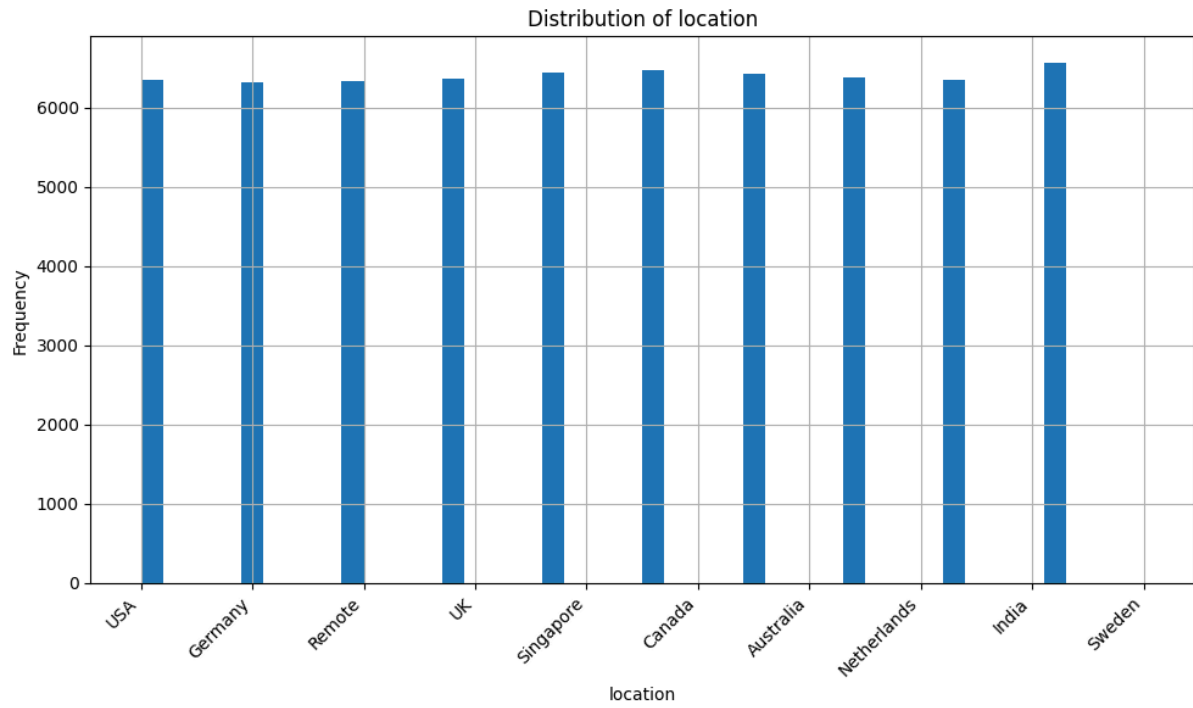
attribute categorie:

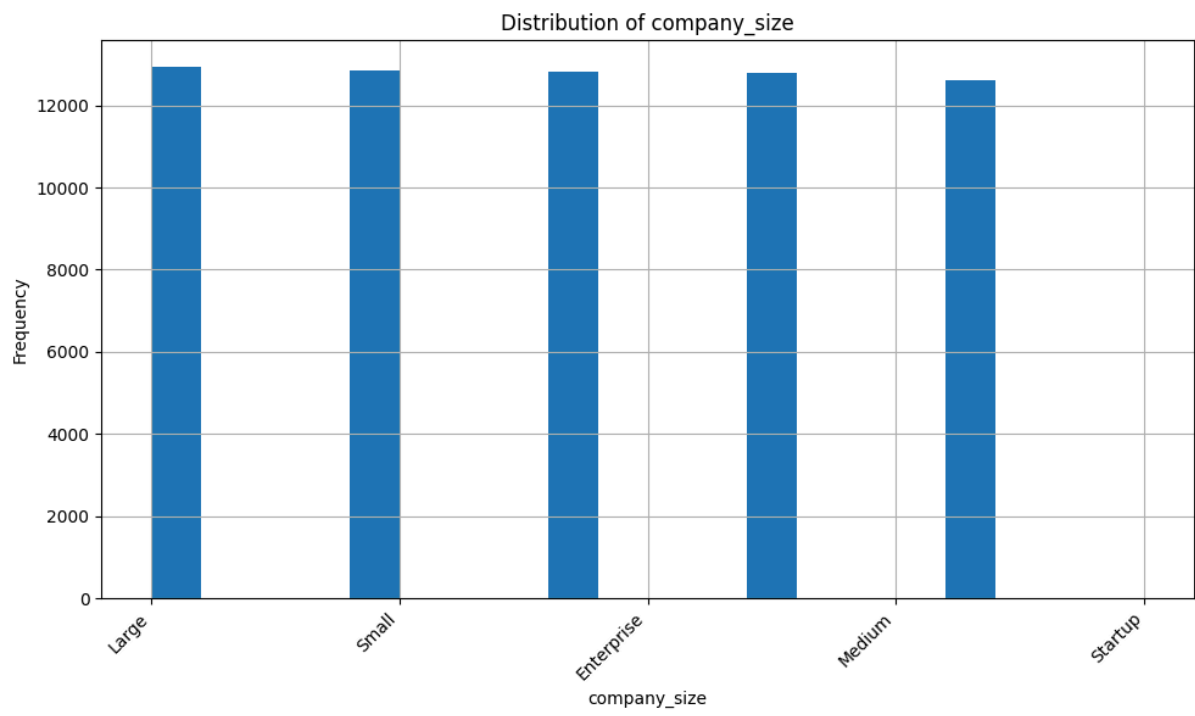
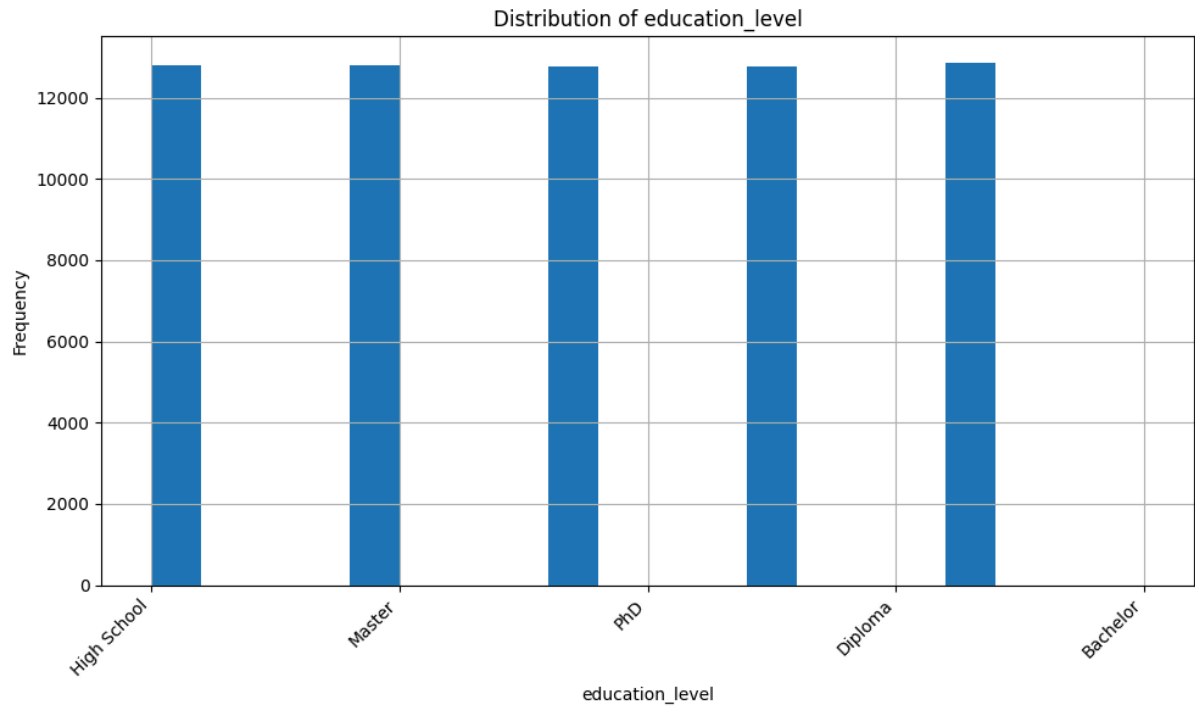
	count	unique
job_title	64000	12
industry	64000	10
remote_work	44847	3
location	64000	10

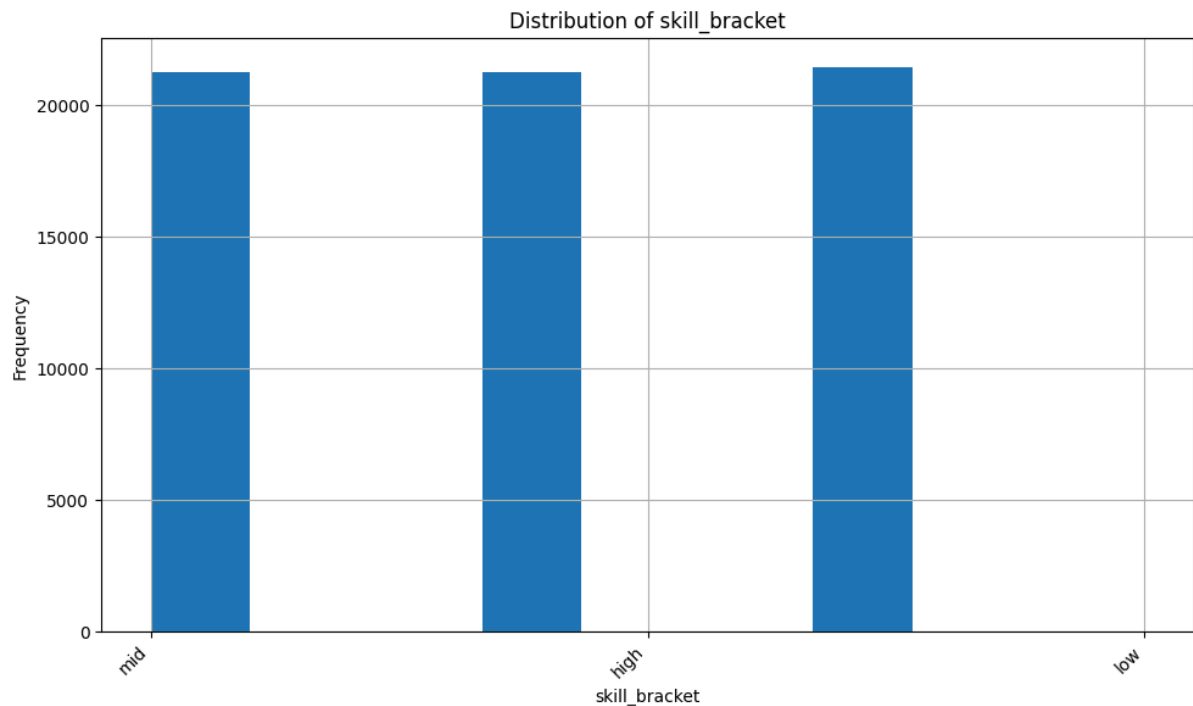
vacation	6400 0	4
education_level	6400 0	5
company_size	6400 0	5
skill_bracket	6400 0	3



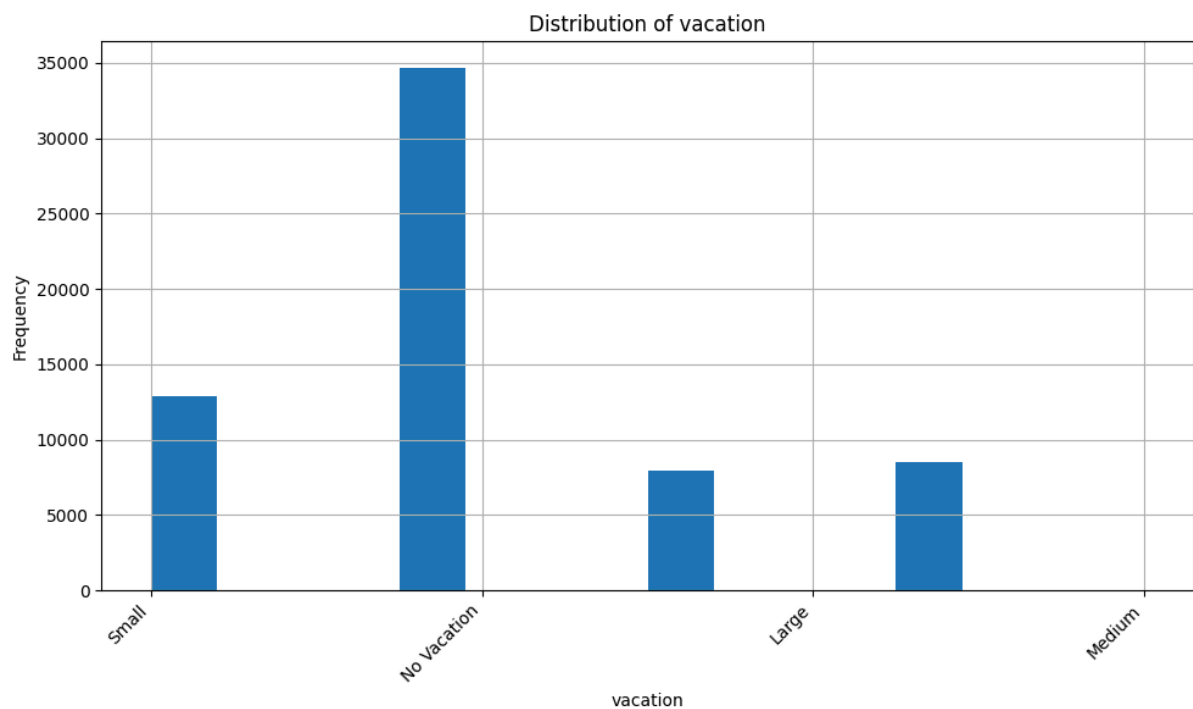








Dezechilibru intre clase:

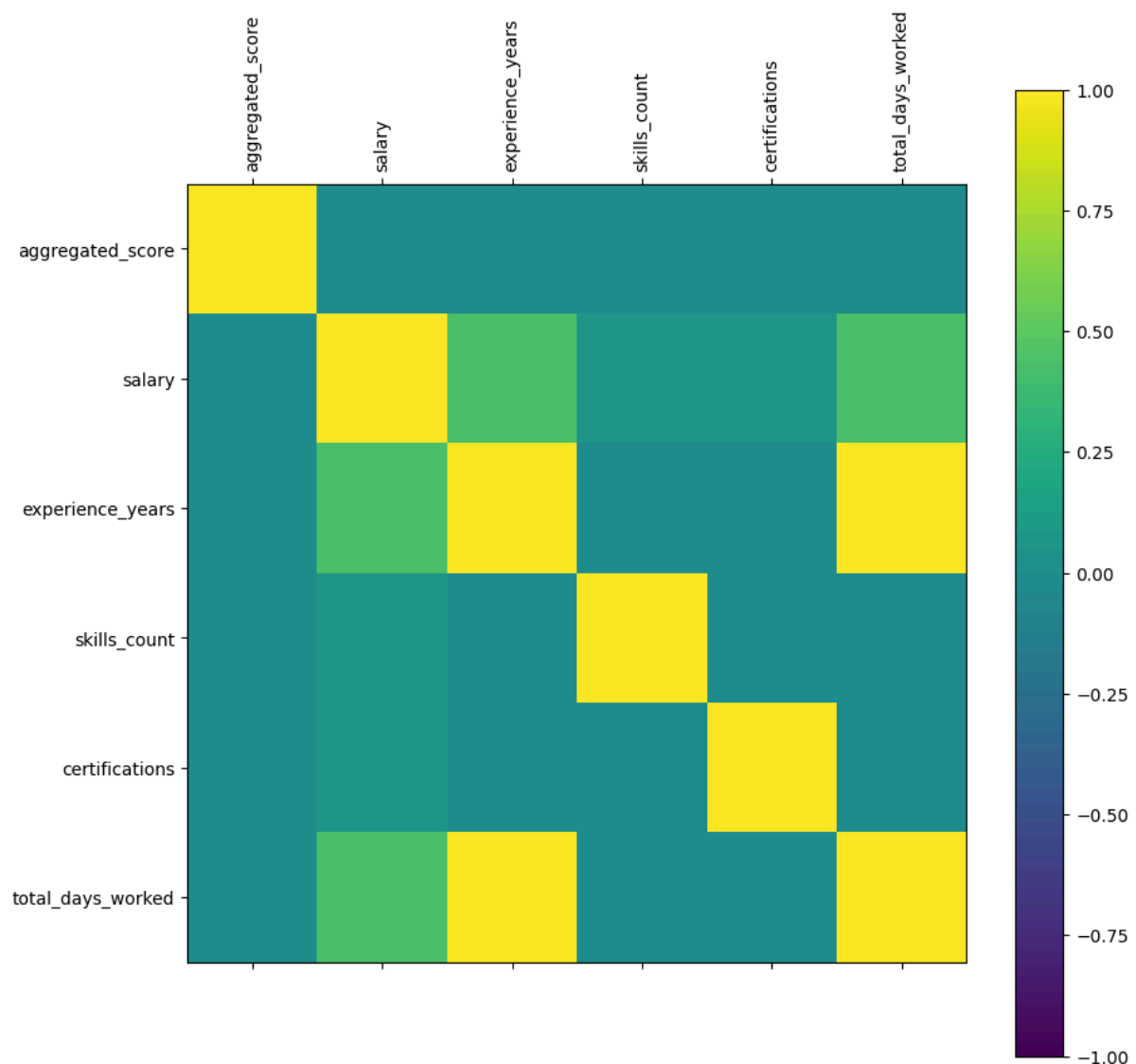


Se observa dezechilibrul dintre numarul de exemple din fiecare clasa. Modelul va distinge cu precizie mare exemplele din “No Vacation” dar va avea probleme la restul. Metricile precision,

recall si F1 sunt importante pentru a evalua performanta
modeului atunci cand clasele sunt dezechilibrate.

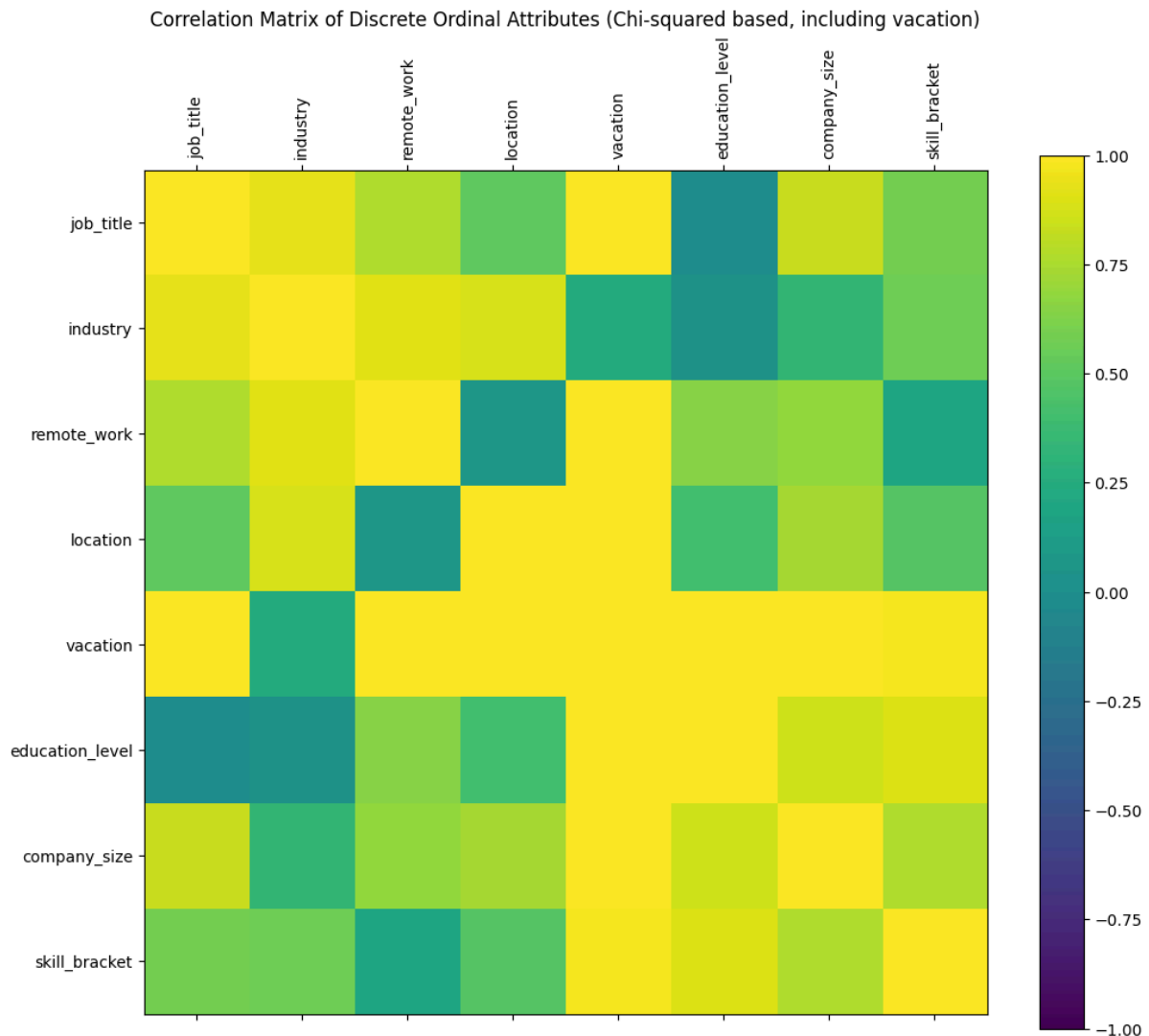
Corelatie intre attribute:

Attribute numerice:



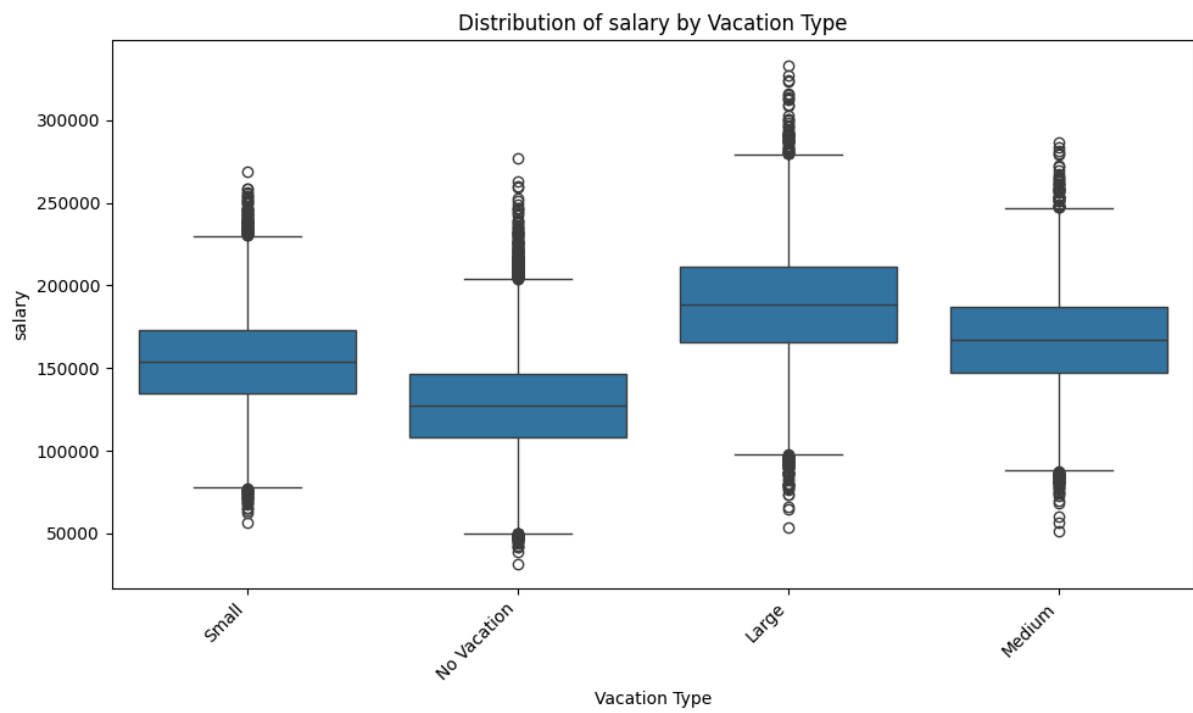
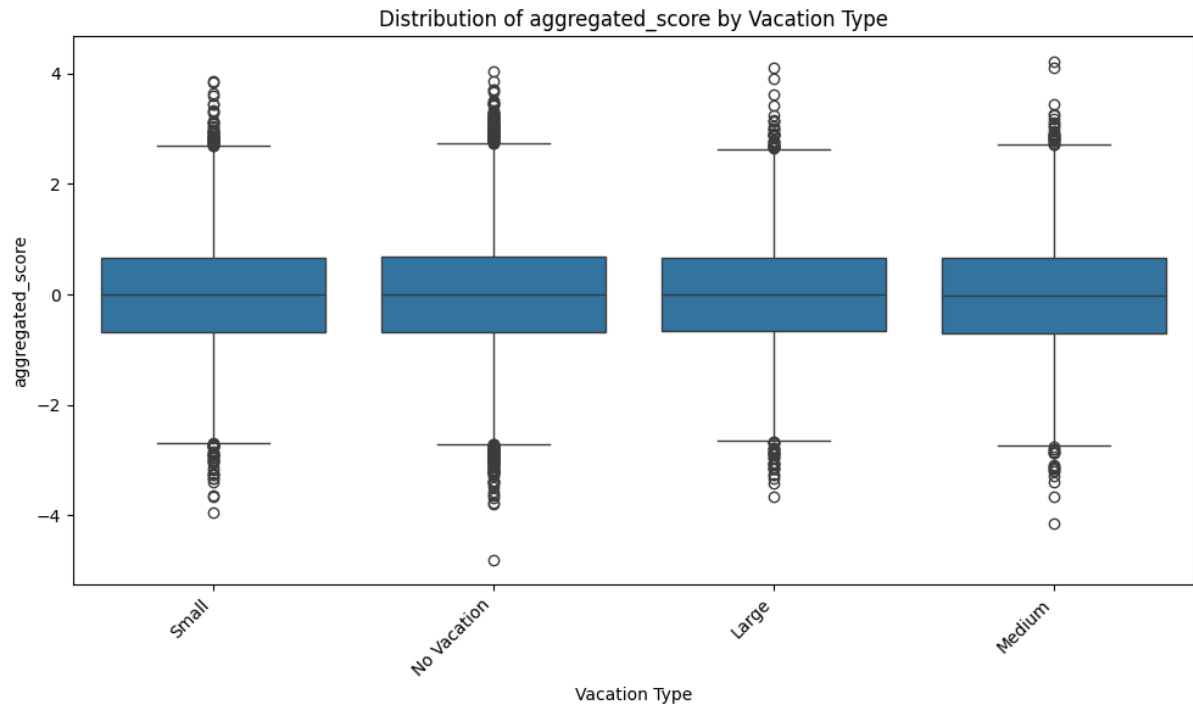
Se observa un grad de corelare ridicat intre total_days_worked
si experience_years cea ce este si normal avand in vedere
semnificatia lor.

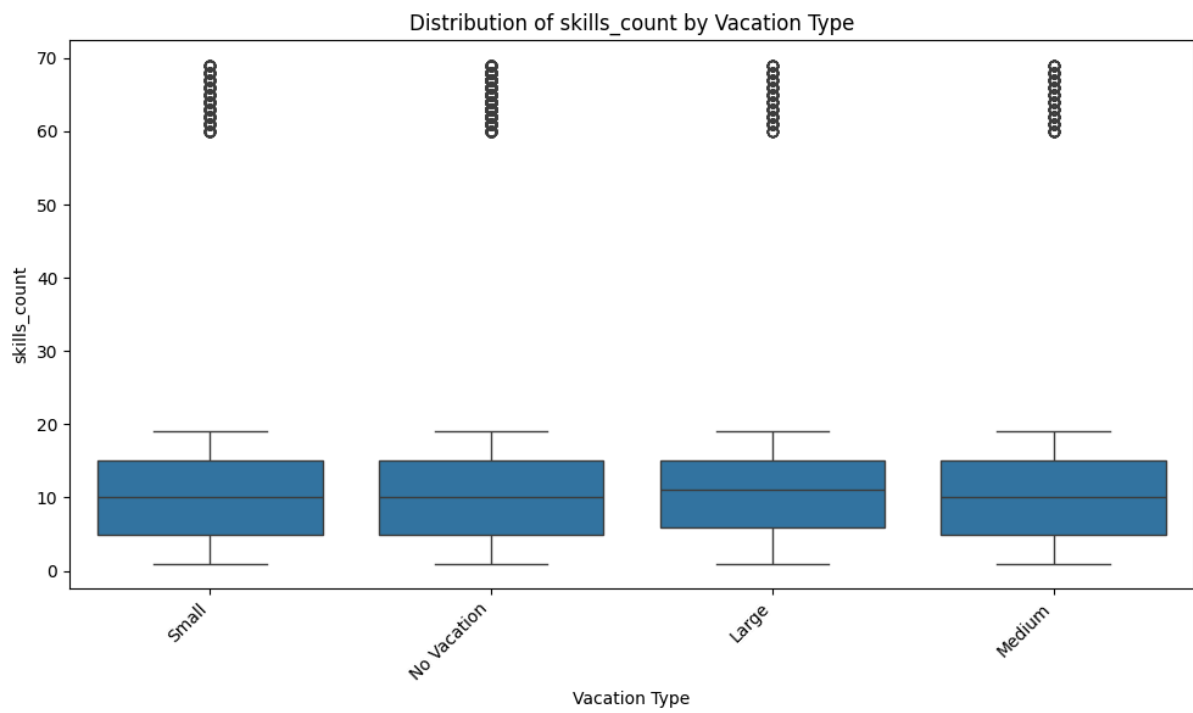
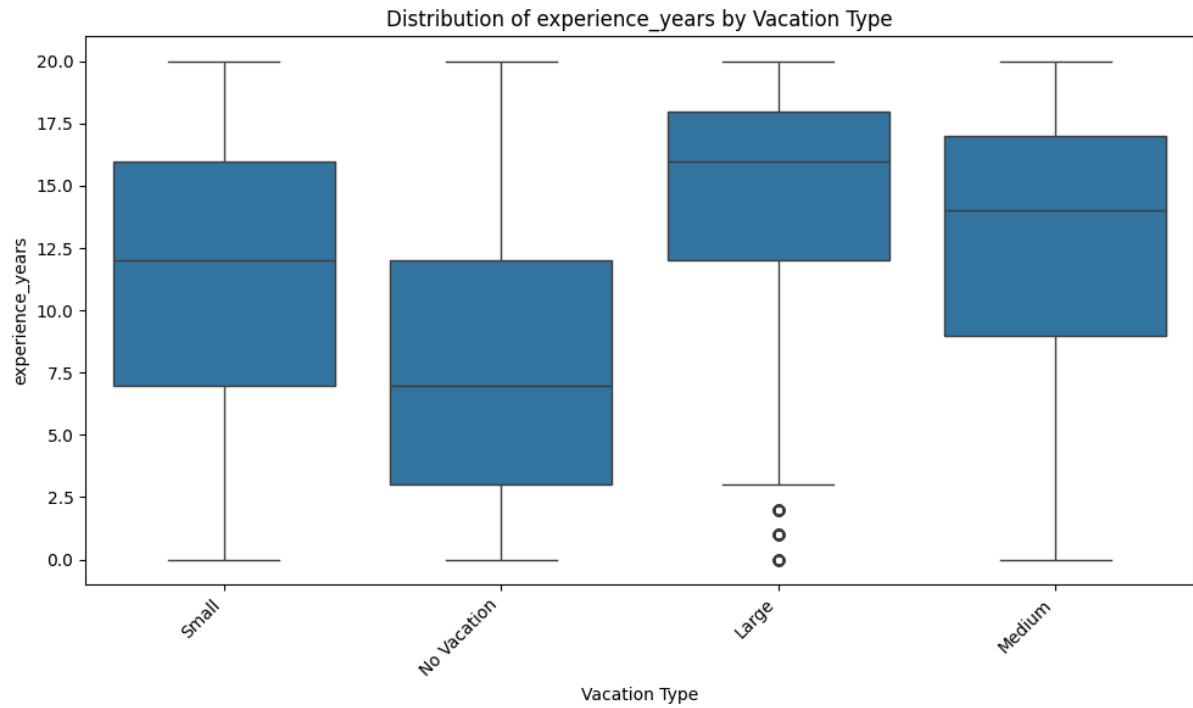
Atribute categorice:

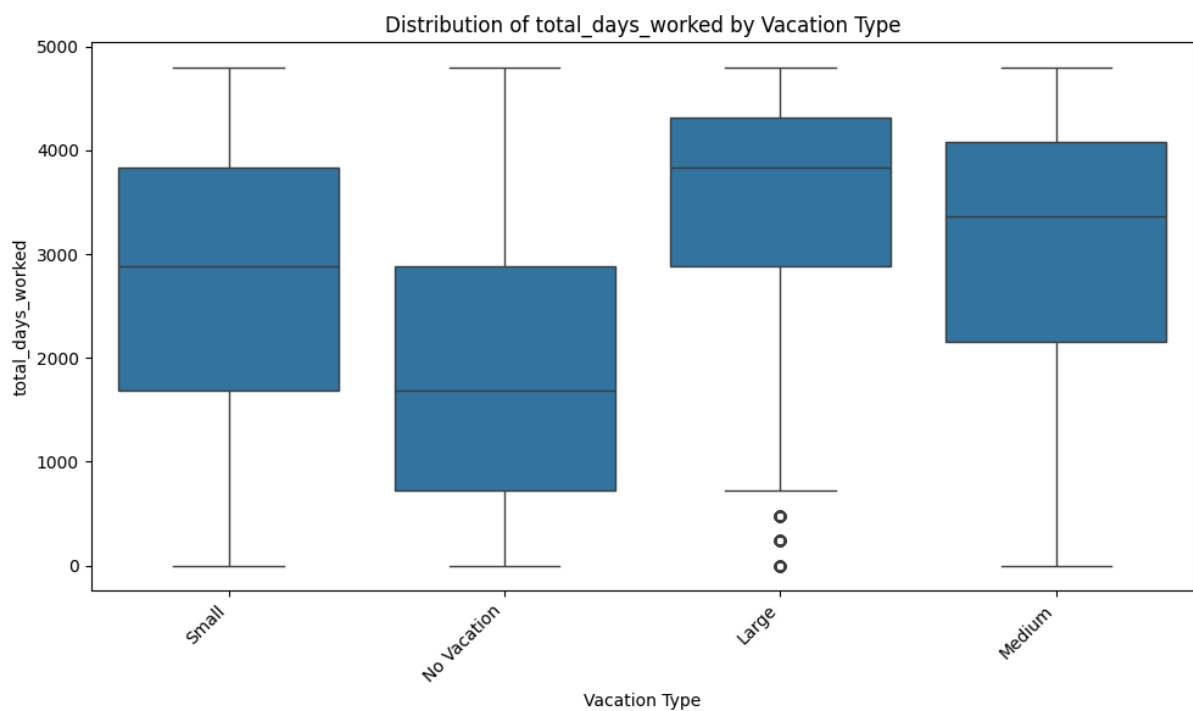
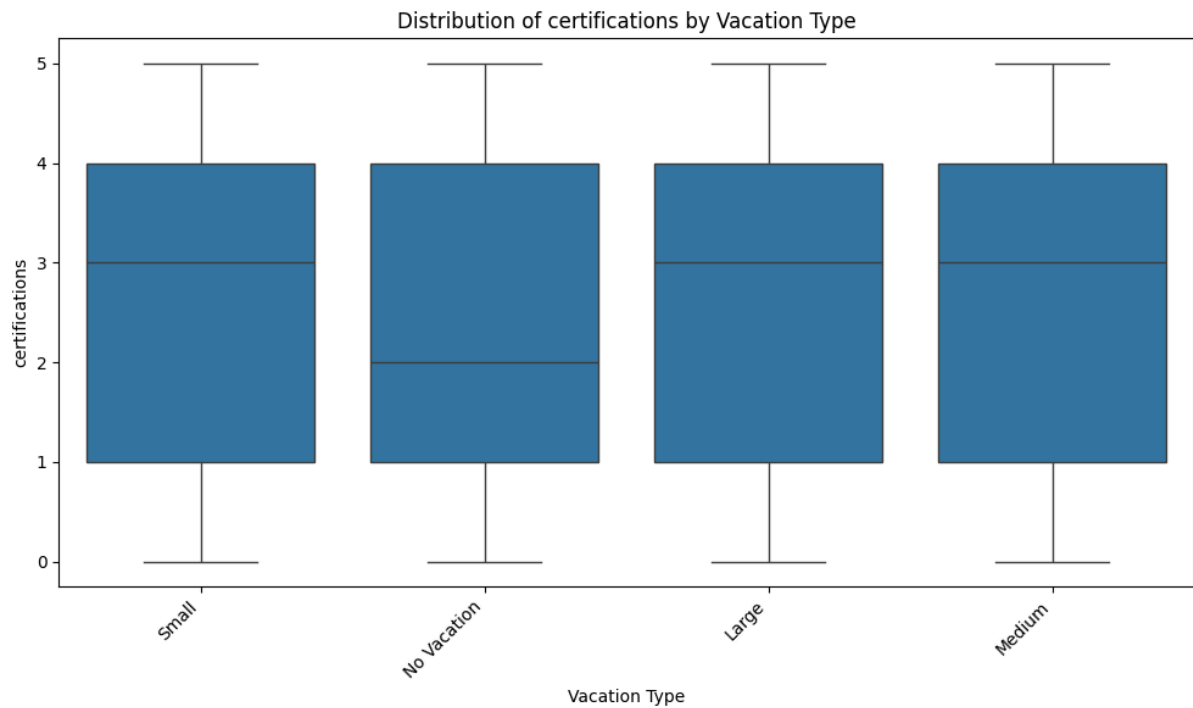


Se observa ca industry si job_title sunt relativ puternic corelate, la fel si skill_bracket cu education_level. De asemenea, in aceasta matrice ies in evidenta attributele care influenteaza semnificativ prezicerea clasei vacation. Se observa ca atributul industry nu este relevant deci in testarea modelului se va putea renunta la el.

Corelatie attribute numerice - sarcina de clasificare:







Se observa grad de corelatie mare intre experience_years - vacation si total_days_worked - vacation.

Corelatie attribute categoriale - sarcina de clasificare:

Se vede in matricea de mai sus (cea cu corelatia dintre atr
catoriale).

Corelatie attribute numerice - sarcina regresie:

Se poate vedea in matricea de mai sus (cea cu corelatia dintre atr numerice).

Corelatia atributelor categorice - sarcina regresiei:

|||||

2. Procesarea datelor:

Doar din `remote_work` lipsesc valori. Pe acestea le voi imputa cu valoarea medie a acestui atribut. Am eliminat outliers si am scalat valorile.

	Etapa de Preprocesare	Ac ur at e ț e	F1-Score (weighted avg)
	Imputare Categoricală	0.67000	0.670995

	Outlier Handling și Imputare Numerică	0.6 67 56 2	0.668356
	Complet Preprocesat (Imputare, Outlier Handlin...	0.6 74 76 6	0.676547

In continuare vom folosi doar setul de date cu imputare categorica avand in vedere ca aduce cea mai buna performanta (chiar daca diferenta este foarte mica)

Clasificare:

Am folosit DecisionTreeClassifier. Prima oara am aplicat algoritmul fara hiperparametrii:

--- Rulare cu parametri impliciti ---

Accuracy: 0.6700

F1-Score (weighted avg): 0.6710

	Parametru	Acuratețe	F1-Score (weighted avg)
0	Implicit	0.67	0.671011

In continuare, am variat hiperparametrul min_sample_leafs (daca intr-o frunza sunt mai putin de min_sample_leafs

exemple, taietura nu mai are loc si se revine la starea precedenta care devine frunza).

Sumar al performanței cu diferite min_samples_leaf:

	min_samples _leaf	Acurate țe	F1-Score (weighted avg)
0	2	0.67398 4	0.673045
1	5	0.70984 4	0.706555
2	10	0.72656 2	0.721773
3	12	0.72742 2	0.723072
4	15	0.73179 7	0.727488
5	20	0.73601 6	0.731929
6	25	0.73929 7	0.734423
7	30	0.74171 9	0.737390
8	40	0.74578 1	0.741709
9	50	0.74445 3	0.739864

1 0	100	0.74164 1	0.737381
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Se observa ca cel mai bun rezultat l-am obtinut cu min_samples_leaf = 40.

Urmatorul hiperparametru folosit este min_samples_split (daca sunt mai putin de min_samples_split exemple intr-o stare, aceasta devine frunza).

Sumar al performanței cu diferite min_samples_split:

	min_samples_split	Acuratețe	F1-Score (weighted avg)
0	60	0.72703 1	0.724339
1	70	0.73289 1	0.729050
2	100	0.74093 8	0.737056
3	120	0.74421 9	0.739859
4	150	0.74710 9	0.742554
5	200	0.74484 4	0.739786

Cel mai bun rezultat cu min_samples_split = 150.

Urmatorul hiperparametru folosit este max_depth (adancimea maxima a arborelui/ nr max de attribute pe care le foloseste)

Sumar al performanței cu diferite max_depth:

	max_depth	Acuratețe	F1-Score (weighted avg)
0	5	0.670312	0.641089
1	8	0.732563	0.730673
2	10	0.737812	0.734822
3	15	0.700250	0.697673
4	20	0.676813	0.676852

Cel mai bun max_depth = 10

Urmatorul hiperparametru class_weight = 'balanced' (ajuta la detectarea claselor minoritare, cum sunt small si medium)

--- Rulare cu class_weight = 'balanced' ---

Accuracy: 0.6658

	Parametru	Acurate țe	F1-Score (weighted avg)
0	class_weight = balanced	0.66578 1	0.667364

Urmatorul hiperparametru SMOTE (genereaza exemple sintetice pentru clasa minoritara)

	Parametru	Acurate țe	F1-Score (weighted avg)
0	SMOTE aplicat	0.67101 6	0.676066

In continuare, am combinat cele mai bune rezultate de la fiecare hiperparametru:

```
min_samples_split = 150
min_samples_leaf = 40
max_depth = 10
class_weight = 'balanced'
SMOTE
```

```
--- Rulare cu min_samples_split = 150, min_samples_leaf = 40,
max_depth = 10, class_weight = 'balanced' și SMOTE ---
Accuracy: 0.7421
```

	Parametru	Acurat ețe	F1-Score (weighted avg)
0	SMOTE + class_weight='balanced' + min_samples_...	0.74212 5	0.743968

Se observa o imbunatatire de aproape 8% a acuratetii si a F1-score ului fata de varianta fara hiperparametrii, insa class_weight si SMOTE nu au adus imbunatatiri semnificative (desi asa ma asteptam) asa ca am reununtat la ele.

In continuare, am incercat si eliminarea anumitor attribute din setul de date.

Am eliminat coloana industry pt ca aceasta nu era aproape deloc relevanta pentru precizia clasei "vacation".

Am rulat cu urmatoorii parametrii:

min_samples_split = 150

min_samples_leaf = 40

fara coloana industry

Original number of features: 54

Number of features after dropping 'industry': 44

--- Rulare cu min_samples_split = 150, min_samples_leaf = 40
și fără coloana industry ---

Accuracy: 0.7494

Classification Report:

precision recall f1-score support

Large 0.77 0.74 0.76 1977

Medium	0.52	0.48	0.50	2132
No Vacation	0.86	0.92	0.89	8672
Small	0.53	0.48	0.50	3219

accuracy		0.75	16000
macro avg	0.67	0.66	0.66 16000
weighted avg	0.74	0.75	0.74 16000

	Parametru	Acuratețe	F1-Score (weighted avg)
0	min_samples_split=150, min_samples_leaf=40, fă...	0.749437	0.743322

Deja se observa o mica imbunatatire (0.007) fata de rularile anterioare.

In continuare, am eliminat si coloana total_days_worked, pentru ca aceasta este puternic corelata cu coloana experience_years, deci este redundanta.

Am rulat cu urmatoorii parametrii:

```
min_samples_split = 150
min_samples_leaf = 40
fara coloana industry
fara coloana total_days_worked
```

Original number of features: 54

Number of features after dropping 'industry' and 'total_days_worked': 42

--- Rulare cu min_samples_split = 150, min_samples_leaf = 40
și fără coloanele industry și total_days_worked ---

Accuracy: 0.7500

Classification Report:

	precision	recall	f1-score	support
Large	0.77	0.71	0.74	1977
Medium	0.50	0.47	0.49	2132
No Vacation	0.87	0.91	0.89	8672
Small	0.54	0.53	0.53	3219
accuracy		0.75		16000
macro avg	0.67	0.65	0.66	16000
weighted avg	0.74	0.75	0.75	16000

	Parametru	Acuratete	F1-Score (weighted avg)
0	min_samples_split=150, min_samples_leaf=40, fă...	0.75	0.746628

! Cel mai bun rezultat obtinut:

Acuratete = 0.75

F1-Score = 0.746

Regresie:

Regresia liniara simpla:

--- Antrenare model de Regresie Liniară ---

--- Rezultate Regresie Liniară Salariu ---

Mean Absolute Error (MAE): 5751.33
Mean Squared Error (MSE): 55247727.07
Root Mean Squared Error (RMSE): 7432.88
R-squared (R²): 0.9600



Deja avem o performanta destul de buna (R-squared: 0.9600)

Am incercat si regresie ridge pentru alpha = 1, 1.25, 1.5, 1.75

--- Antrenare model de Regresie Ridge ---

--- Rezultate Regresie Ridge Salariu ---

Mean Absolute Error (MAE): 5751.28

Mean Squared Error (MSE): 55247881.43

Root Mean Squared Error (RMSE): 7432.89

R-squared (R2): 0.9600



--- Antrenare model de Regresie Ridge cu $\alpha = 1.25$ ---

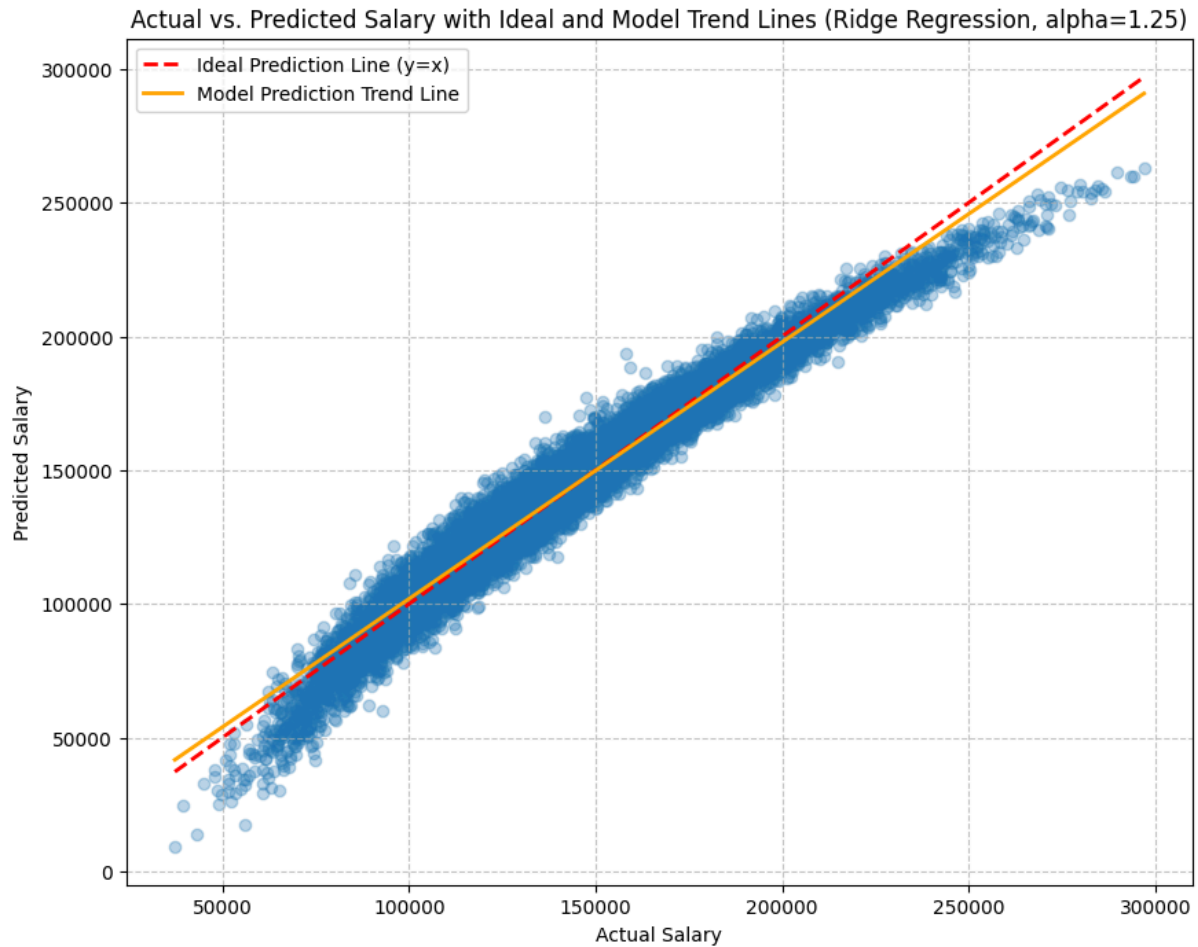
--- Rezultate Regresie Ridge Salariu cu $\alpha = 1.25$ ---

Mean Absolute Error (MAE): 5751.28

Mean Squared Error (MSE): 55248014.94

Root Mean Squared Error (RMSE): 7432.90

R-squared (R2): 0.9600



--- Antrenare model de Regresie Ridge cu $\alpha = 1.5$ ---

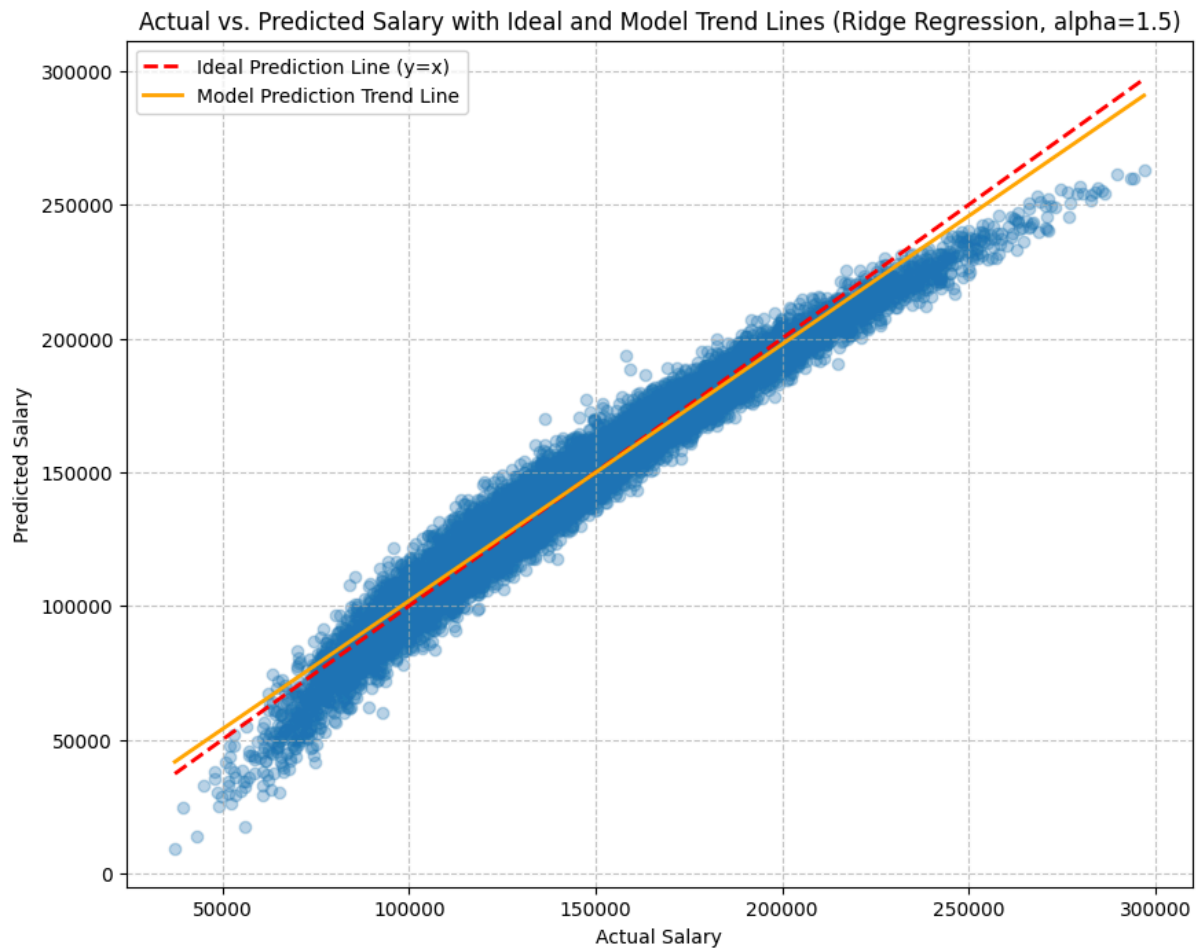
--- Rezultate Regresie Ridge Salariu cu $\alpha = 1.5$ ---

Mean Absolute Error (MAE): 5751.28

Mean Squared Error (MSE): 55248151.14

Root Mean Squared Error (RMSE): 7432.91

R-squared (R^2): 0.9600



--- Antrenare model de Regresie Ridge cu $\alpha = 1.75$ ---

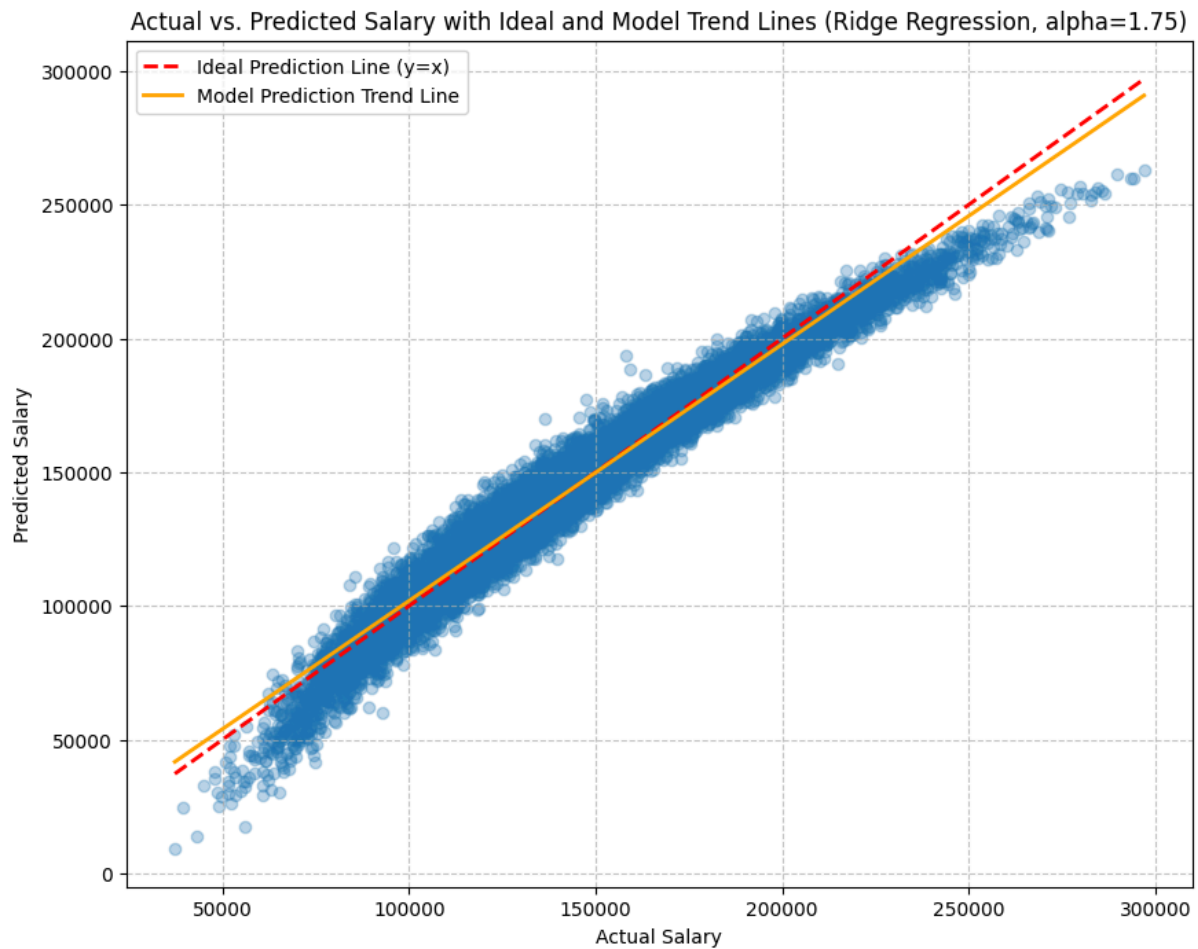
--- Rezultate Regresie Ridge Salariu cu $\alpha = 1.75$ ---

Mean Absolute Error (MAE): 5751.27

Mean Squared Error (MSE): 55248290.01

Root Mean Squared Error (RMSE): 7432.92

R-squared (R^2): 0.9600



Sumarul performanței modelelor de regresie:

	Model	MAE	MSE	RMSE	R-squared
0	Linear Regression	5751.325273	5.524773e+07	7432.881478	0.960005
1	Ridge Regression (alpha=1.0)	5751.282223	5.524788e+07	7432.891862	0.960005
2	Ridge Regression (alpha=1.25)	5751.279163	5.524801e+07	7432.900843	0.960005

3	Ridge Regression (alpha=1.5)	5751.27 6476	5.524815 e+07	7432.91 0005	0.9600 05
4	Ridge Regression (alpha=1.75)	5751.27 3938	5.524829 e+07	7432.91 9347	0.9600 05

Regresia ridge nu a adus nicio imbunatatire fata de cea normala.

Am incercat si regresia polinomiala de gradul 2:

--- Antrenare model de Regresie Polinomială (grad 2) ---

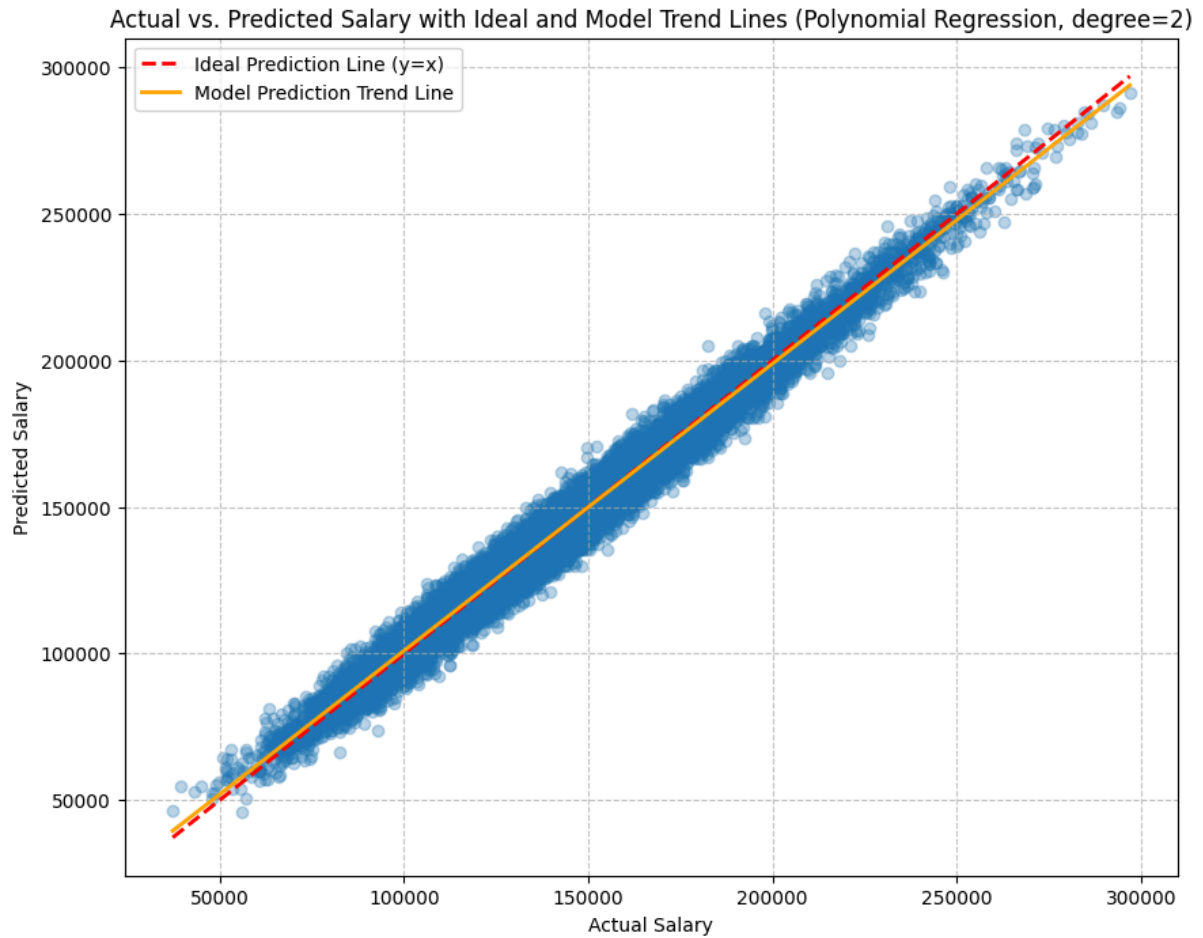
--- Rezultate Regresie Polinomială Salariu (grad 2) ---

Mean Absolute Error (MAE): 4242.16

Mean Squared Error (MSE): 28242537.25

Root Mean Squared Error (RMSE): 5314.37

R-squared (R2): 0.9796



R-squared a crescut semnificativ si se poate vedea si pe grafic imbunatatirea.

!Cea mai buna varianta